Lesson 4 Teacher's Key

TIC-80

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

Fixed enemy state declared once at the top of the code, plus one new branch inside the existing bullets loop:

```
enemy_alive=true
ex,ey=60,10
bullets={}

function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
  table.insert(bullets,{bx=x,by=y})
 end
 for i=#bullets,1,-1 do
  local b=bullets[i]
  b.by=b.by-2
  if enemy_alive and b.bx<ex+8 and b.bx+8>ex and b.by<ey+8 and b.by+8>ey
then
   enemy_alive=false
   table.remove(bullets,i)
  elseif b.by<0 then
   table.remove(bullets,i)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 if enemy_alive then spr(3,ex,ey,0) end
 spr(1,x,y,0)
end
```

Level 2 - Complete Worked Solution

The enemy_alive check and the overlap check together mean that once enemy_alive is false, that branch can never run again for any later bullet - so the enemy can only ever be "destroyed" once, even if two bullets happen to overlap it on the very same frame. No extra code is needed for Level 2.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on hit

Design a short impact sound in the SFX Editor (F4) in an empty slot (e.g. slot 2), then call sfx(2) right where enemy_alive is set to false:

```
 if enemy_alive and b.bx<ex+8 and b.bx+8>ex and b.by<ey+8 and b.by+8>ey
then
  enemy_alive=false
  sfx(2)
  table.remove(bullets,i)
```

Visual: give the enemy its own look

Draw enemy sprite index 3 as visually distinct from the player and bullet (different shape or color) in the Sprite Editor - this alone satisfies most of "its own look" without any extra code, the same way Lesson 3's bullet used its own sprite index.

Structure: keep enemy state declared once

Keep enemy_alive, ex, and ey declared once at the top of the code, and only ever change enemy_alive from inside the collision check - resist checking or setting it from more than one place, even though everything still technically lives inside the single TIC() function.

Complete Level 3 solution (all 3 goals combined)

Everything above combined into the one TIC() function:

```
enemy_alive=true
ex,ey=60,10
bullets={}

function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
  table.insert(bullets,{bx=x,by=y})
 end
 for i=#bullets,1,-1 do
  local b=bullets[i]
  b.by=b.by-2
  if enemy_alive and b.bx<ex+8 and b.bx+8>ex and b.by<ey+8 and b.by+8>ey
then
   enemy_alive=false
   sfx(2)
   table.remove(bullets,i)
  elseif b.by<0 then
   table.remove(bullets,i)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 if enemy_alive then spr(3,ex,ey,0) end
```

```
 spr(1,x,y,0)
end
```